Marginal distribution, 26-256 PSFM

1 Introduction

$$\Pr_\theta(H_t = h) = \sum_{g'=0}^{\infty} \int_{\mathbb{R}} \pi_t(g', z, u|\theta) \sum_{n=0}^{\infty} e^{-\lambda_{\text{seed}}} \frac{\lambda^{n_{\text{seed}}}}{n!} \int_{\Delta_{m-1}} [s^h] \prod_{i=1}^{m} (1 - q_i + q_i s) \operatorname{Dir}(d\mathbf{p} : \alpha \mathbf{1}_m) dz\, du \tag{1}$$

$$q_i = \begin{cases} 0, & E_i < b, \\ \left(1 - e^{-[r_H(E_i) + r_0]}\right) \dfrac{r_H(E_i)}{r_H(E_i) + r_0}, & E_i \geq b, \end{cases} \quad \begin{aligned} r_H(E) &= -\log(1 - P_H) E^{\beta_E}, \quad P_H \sim \text{Beta}(\alpha_H, \beta_H), \\ r_0 &= -\log(1 - P_D), \qquad\quad P_D \sim \text{Beta}(\alpha_D, \beta_D). \end{aligned} \tag{2}$$

$$\left.\begin{aligned} u_t &= \phi_W u_{t-1} + \sigma_W \sqrt{1 - \phi_W^2}\, \eta_t, \; W_t = w_{\text{total}} e^{u_t} \\ Z_t &= \phi_Z Z_{t-1} + \sqrt{1 - \phi_Z^2}\, \eta_t^2 = \rho_{WZ}\, \eta_t^W + \sqrt{1 - \rho_{\text{WZ}}^2 + \xi_t} \end{aligned}\right\} \eta_t^W \sim \mathcal{N}(0, 1), \; \xi_t \sim \mathcal{N}(0, 1 \tag{3}$$

$$k(z) = \text{round}_{\mathbb{Z}}(K_0 e^{\beta_{kz}}), \; m = \min\{g' + n, k(z)\} \tag{4}$$

$$E_i = w_{\text{total}} e^u p_i \quad n \sim \text{Poisson}(\lambda_{\text{seed}}), \quad p_i \in \mathbf{p} \sim \text{Dirichlet}(\alpha \mathbf{1}_m) \tag{5}$$